

Teaching Exam

Subject – Computer Science

Topic – Computer Network

Lecture No.- 11



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Topics to be Covered

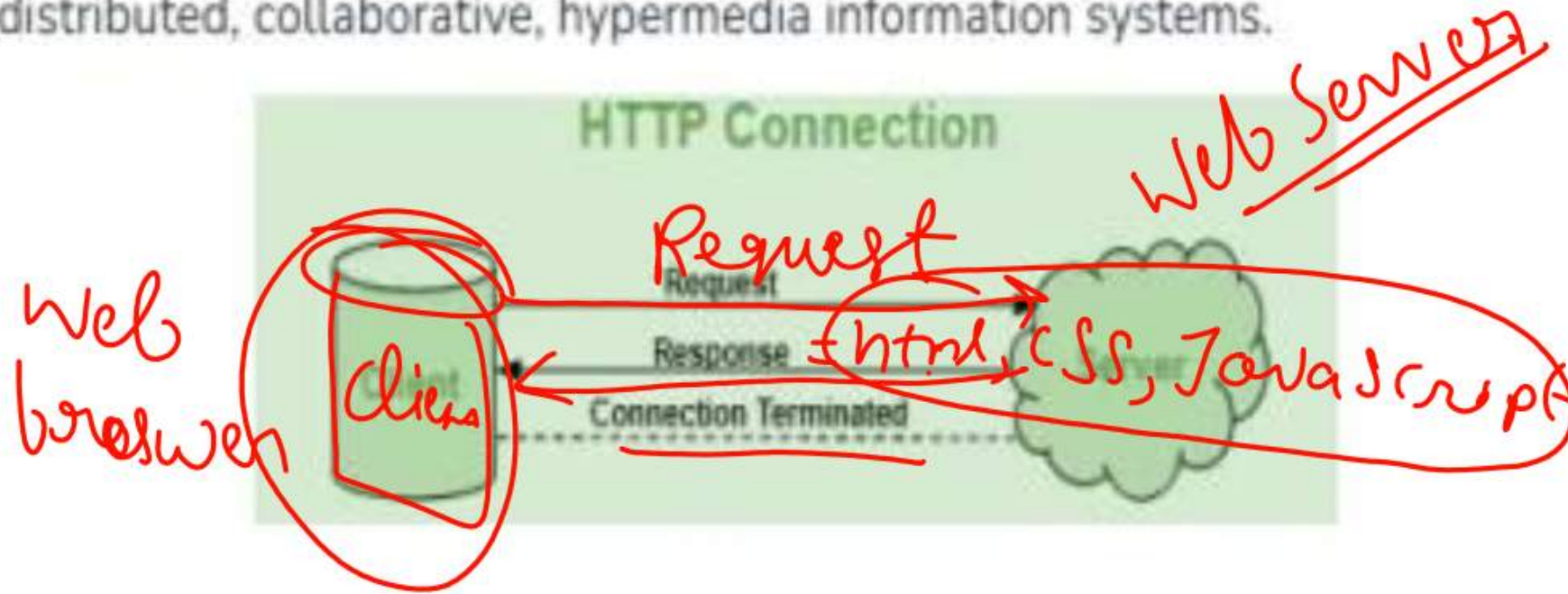


Topic

- 1:- HTTP PROTOCOL ✓
- 2:- HTTPS PROTOCOL ✓
- 3:- Difference between HTTP & HTTPS ✓
- 4:- WWW ✓
- 5:- HTML ✓
- 6:- XML ✓
- 7:- DOMAIN NAME, URL, WEB BROWSER, WEB SERVER, WEB HOSTING

HTTP (Hypertext Transfer Protocol) is a fundamental protocol of the Internet, enabling the transfer of data between a client and a server. It is the foundation of data communication for the World Wide Web.

HTTP provides a standard between a web browser and a web server to establish communication. It is a set of rules for transferring data from one computer to another. Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP. It is an application protocol that is used for distributed, collaborative, hypermedia information systems.



Methods of HTTP

- **GET**: Used to retrieve data from a specified resource. It should have no side effects and is commonly used for fetching web pages, images, etc.
- **POST**: Used to submit data to be processed by a specified resource. It is suitable for form submissions, file uploads, and creating new resources.
- **PUT**: Used to update or create a resource on the server. It replaces the entire resource with the data provided in the request body.
- **PATCH**: Similar to PUT but used for partial modifications to a resource. It updates specific fields of a resource rather than replacing the entire resource.
- **DELETE**: Used to remove a specified resource from the server.

POSTMAN

GET: → To retrieve or fetching Web-Pages (html pages)
Photos(media) from the Server

POST: → To Submit the data On your application
to Server

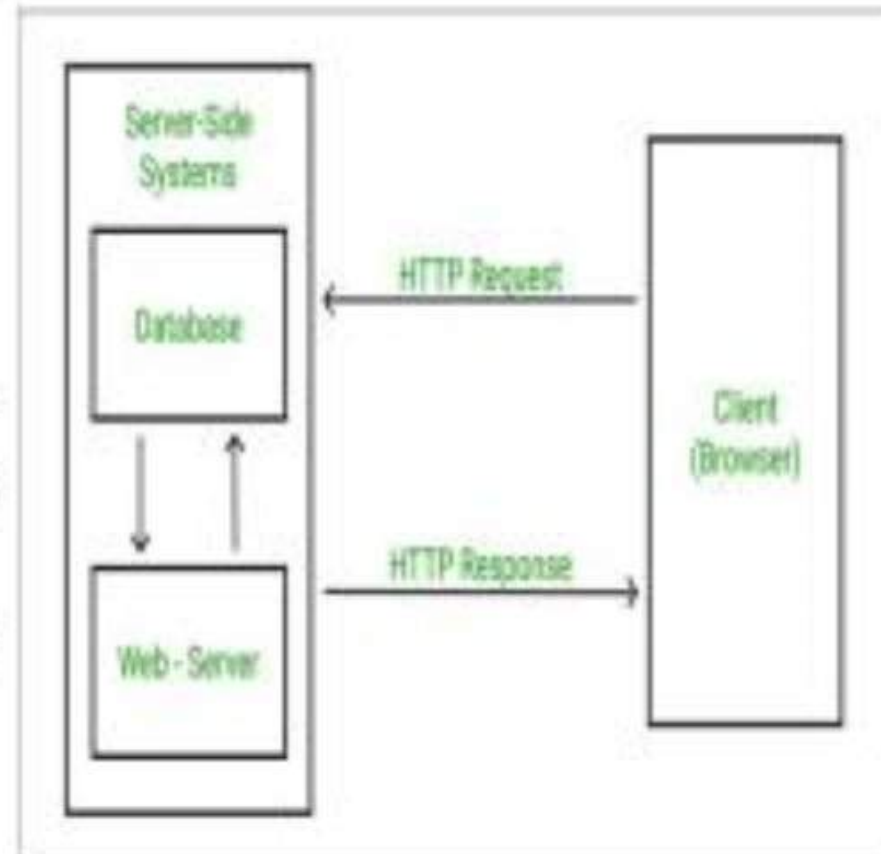
Put: → To ^{complete} update our resource (application) On Server

Patch → To Partially update ; , , ,

HTTP Request/Response:

HTTP is a request-response protocol, which means that for every request sent by a client (typically a web browser), the server responds with a corresponding response. The basic flow of an HTTP request-response cycle is as follows:

- **Client sends an HTTP request:** The client (usually a web browser) initiates the process by sending an HTTP request to the server. This request includes a request method (GET, POST, PUT, DELETE, etc.), the target URI (Uniform Resource Identifier, e.g., a URL), headers, and an optional request body.
- **Server processes the request:** The server receives the request and processes it based on the requested method and resource. This may involve retrieving data from a database, executing server-side scripts, or performing other operations.
- **Server sends an HTTP response:** After processing the request, the server sends an HTTP response back to the client. The response includes a status code (e.g., 200 OK, 404 Not Found), response headers, and an optional response body containing the requested data or content.
- **Client processes the response:** The client receives the server's response and processes it accordingly. For example, if the response contains an HTML page, the browser will render and display it. If it's an image or other media file, the browser will display or handle it appropriately.



Topic- HTTP Protocol



TEACHING
WALLAH

Sender (client/
Header → Domain
Name

HTTP
Request

Concept	Description
HTTP Request Headers	Text information in key-value pairs sent with every HTTP request, conveying details such as client's browser, requested data, and more.
HTTP Request Body	Contains data being submitted to the web server, such as form inputs, authentication details, or other payload data.
HTTP Response	Sent by the server in answer to an HTTP request, consisting of an HTTP status code, response headers, and optional response body.
HTTP Status Codes	3-digit codes indicating the outcome of an HTTP request, categorized into 1xx (Informational), 2xx (Success), 3xx (Redirection), 4xx (Client Error), and 5xx (Server Error) blocks.
HTTP Response Headers	Provide important information like data format, language, and content type in the response, helping clients interpret the data correctly.
HTTP Response Body	Contains the requested information sent by the server, such as HTML content for webpages, JSON data, or other data based on the client's request.

Header
Request Body

HTTP
Response

<head> <title> </title>
</head>
Header (Response)
Response Body
Response Code

Advantages

- Platform independence: Works on any operating system
- Compatibility: Compatible with various protocols and technologies
- Efficiency: Optimized for performance
- Security: Supports encryption for secure data transfer

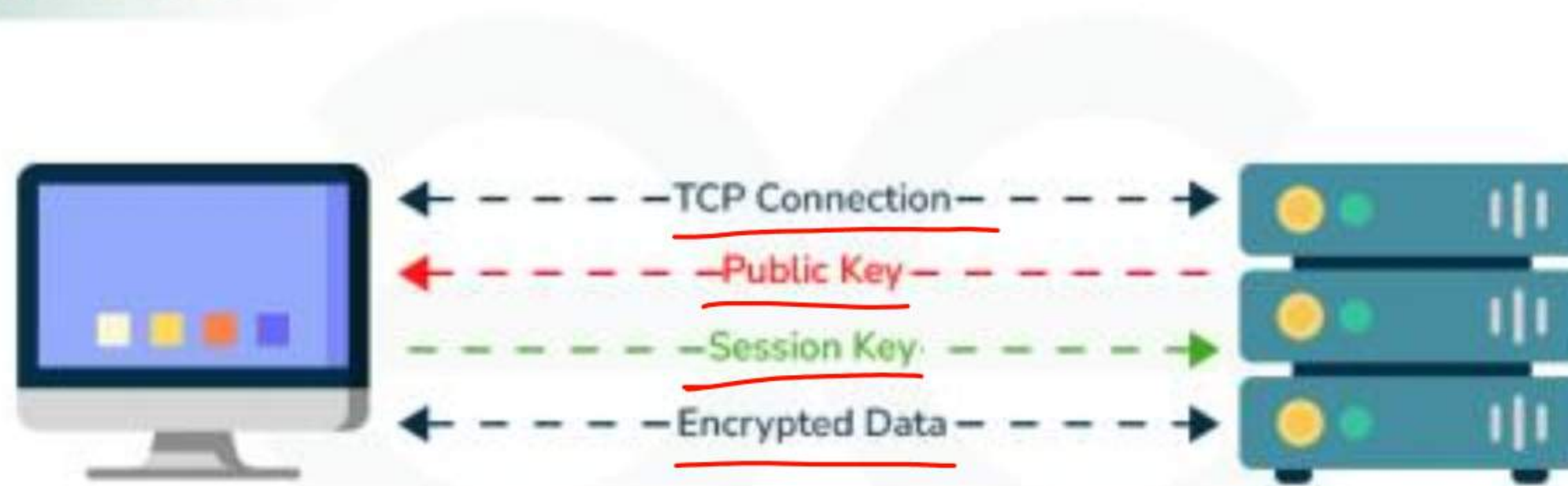
Disadvantages

- Lack of security: Vulnerable to attacks like man in the middle
- Performance issues: Can be slow for large data transfers
- Statelessness: Requires additional mechanisms for maintaining state

HTTPS stands for HyperText Transfer Protocol Secure. It is the most common protocol for sending data between a web browser and a website. HTTPS is the secure variant of HTTP and is used to communicate between the user's browser and the website, ensuring that data transfer is encrypted for added security.

Any website, especially those requiring login details, should use HTTPS. You can see a padlock icon in the URL bar, which means the page is secure. Browsers, like Google Chrome, treat HTTPS seriously and mark non-HTTPS websites as "Not Secure."

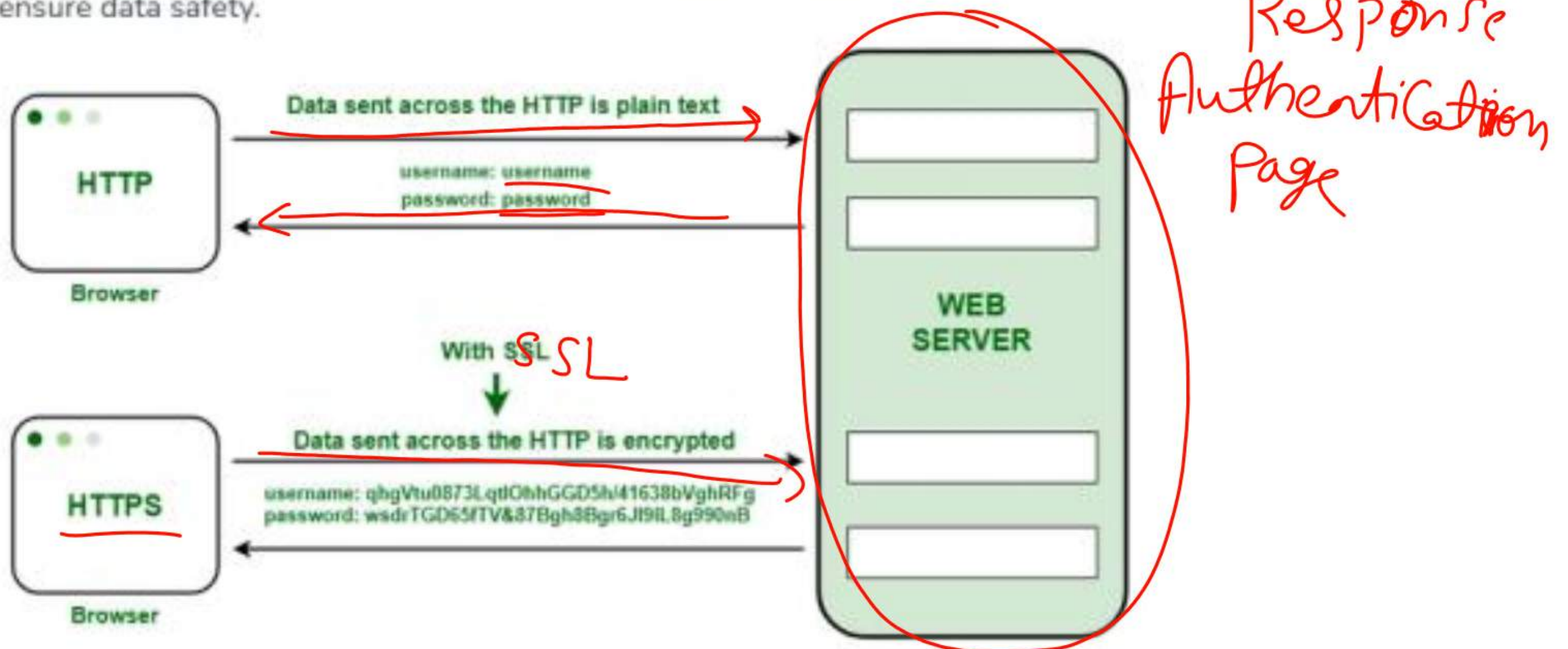
HTTPS



Topic- HTTPS Protocol

HTTPS establishes the communication between the browser and the web server. It uses the **Secure Socket Layer (SSL)** and **Transport Layer Security (TLS)** protocol for establishing communication. The new version of SSL is **TLS(Transport Layer Security)**.

HTTPS uses the conventional HTTP protocol and adds a layer of SSL/TLS over it. The [workflow of HTTP and HTTPS](#) remains the same, the browsers and servers still communicate with each other using the HTTP protocol. However, this is done over a secure SSL connection. The SSL connection is responsible for the encryption and decryption of the data that is being exchanged to ensure data safety.



Why HTTPS Matters and What Happens Without It?

HTTPS is important because it keeps the information on websites safe from being easily viewed or stolen by anyone who might be spying on the network. When a website uses regular HTTP, data is sent in small chunks called packets that can easily be intercepted using free software. This makes communication, especially over public Wi-Fi, very vulnerable to attacks.

On the other hand, HTTPS encrypts the data, so even if someone manages to intercept the packets, they will appear as random, unreadable characters. For example:

- **Before encryption:**

"This is a string of text that is completely readable"

- **After encryption:**

"ITM0IRyiEhVpa6VnKyExMiEgNveroyWBP1gGyfkflYjDaaFf/Kn3bo3OfghBPDWo6AfSH1NtL8N7ITEwIXc1gU5X73xMsJormzzXlwOyrCs+9XCPk63Y+z0="

encrypted data + SSL protocol

Secure Socket Layer (SSL)

The main responsibility of SSL is to ensure that the data transfer between the communicating systems is secure and reliable. It is the standard security technology that is used for encryption and decryption of data during the transmission of requests.

As discussed earlier, HTTPS is basically the same old HTTP but with SSL. For establishing a secure communication link between the communicating devices, SSL uses a digital certificate called SSL certificate.

There are two major roles of the SSL layer

- Ensuring that the browser communicates with the required server directly.
- Ensuring that only the communicating systems have access to the messages they exchange.

Handwritten: $HTTPS = HTTP + \text{SSL/TLS}$ ned

Advantage of HTTPS

- **Secure Communication:** HTTPS establishes a secure communication link between the communicating system by providing encryption during transmission.
- **Data Integrity:** By encrypting the data, HTTPS ensures data integrity. This implies that even if the data is compromised at any point, the hackers won't be able to read or modify the data being exchanged.
- **Privacy and Security:** HTTPS prevents attackers from accessing the data being exchanged passively, thereby protecting the privacy and security of the users.
- **Faster Performance:** TTPS encrypts the data and reduces its size. Smaller size accounts for faster data transmission in the case of HTTPS.

Handwritten: It helps Data in form of Encryption and Decryption With Security

Handwritten: 1er

HTTP vs HTTPS

Below are the basic differences between HTTP and HTTPS.

HTTP	HTTPS
HTTP stands for HyperText Transfer Protocol.	HTTPS stands for HyperText Transfer Protocol <u>Secure</u> .
URL begins with " <u>http://</u> ".	URL starts with " <u>https://</u> ".
HTTP Works at the <u>Application Layer</u> .	HTTPS works at <u>Transport Layer</u> .
HTTP speed is faster than <u>HTTPS</u> .	HTTPS speed is slower than HTTP.

The World Wide Web (WWW), often called the Web, is a system of interconnected webpages and information that you can access using the Internet. It was created to help people share and find information easily, using links that connect different pages together. The Web allows us to browse websites, watch videos, shop online, and connect with others around the world through our computers and phones.

All public websites or web pages that people may access on their local computers and other devices through the internet are collectively known as the World Wide Web or W3. Users can get further information by navigating to links interconnecting these pages and documents. This data may be presented in text, picture, audio, or video formats on the internet.



World Wide Web

Fact: Today, it connects over **63% of the world's population**, making it one of the most powerful tools for communication and information sharing.

Key Parts of the Web

www

The Web has three main building blocks that make it work:

- URL (Uniform Resource Locator): This is the address of a webpage, like https://www.example.com./ It tells your browser exactly where to find the page.
- HTTP (Hypertext Transfer Protocol): This is the set of rules that lets your browser and the server talk to each other to send and receive webpages.
- HTML (Hypertext Markup Language): This is the code that tells browsers how to display a webpage, including where to put text, pictures, and links.

World Wide Web (WWW) Vs Internet

It's easy to mix up the Web and the Internet, but they're different:

Aspect	World Wide Web	Internet
What It Is	A collection of webpages and websites you access with a browser.	A global network connecting computers.
Started	1989 by Tim Berners-Lee at CERN.	1960s as <u>ARPANET</u> .
Purpose	To share and explore information like text, images, and videos.	To connect devices and share data.
How You Use It	Through browsers like Chrome or Firefox.	Through any connected device for email, apps, etc.
Example	Visiting a website like Wikipedia.	Sending an email or streaming a video.

Working of World Wide Web(WWW)

A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal.

The below diagram indicates how the Web operates just like client-server architecture of the internet. When users request web pages or other information, then the web browser of your system request to the server for the information and then the web server provide requested services to web browser back and finally the requested service is utilized by the user who made the request.



HTML stands for **Hyper Text Markup Language**, which is the core language used to structure content on the web. It organizes text, images, links, and media using tags and elements that browsers can interpret. As of 2025, over 95% of websites rely on HTML alongside CSS and JavaScript, making it a fundamental tool in modern web development.

- It is a markup language, not a programming language. This means it annotates text to define how it is structured and displayed by [web browsers](#).
- It is a **static language**, meaning it does not inherently provide interactive features but can be combined with [CSS](#) for styling and [JavaScript](#) for interactivity.

In a nutshell, HTML is all about **organizing and displaying information** on a webpage. We can think of it as the **bones or structure** of a webpage.

Basic HTML Code Example

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Webpage</title>
</head>
<body>
  <h1>Welcome to My Webpage</h1>
  <p>This is my first paragraph of text!</p>
</body>
</html>
```

HTML Page Structure

The basic structure of an HTML page is shown below. It contains the essential building-block elements (i.e. doctype declaration, HTML, head, title, and body elements) upon which all web pages are created.

HTML Page Structure

```
<!DOCTYPE html>    ← Tells version of HTML
<html>              ← HTML Root Element
<head>              ← Used to contain page HTML metadata
  <title>Page Title</title> ← Title of HTML page
</head>
<body>              ← Hold content of HTML
  <h2>Heading Content</h2> ← HTML heading tag
  <p>Paragraph Content</p> ← HTML paragraph tag
</body>
</html>
```

Applications of HTML

- **Web Development:** HTML is the backbone of every webpage. It structures the content and integrates multimedia, hyperlinks, and more.
- Web Applications: HTML, in combination with CSS and JavaScript, powers complex web applications (e.g., Google Docs, Trello).
- Emails: HTML emails use table-based layouts and embedded media to deliver rich, interactive content.
- Mobile App Development: HTML5 is used with frameworks like PhoneGap to build mobile apps for iOS and Android.

Limitations of HTML

- No Logic or Functionality: HTML cannot handle complex logic, interactivity, or dynamic content on its own. It requires JavaScript for such tasks.
- SEO Limitations: While HTML provides structure, it's not enough by itself for search engine optimization (SEO). Proper metadata and content structuring, as well as external SEO practices, are necessary.
- **Limited Styles:** While HTML can handle basic styles via the style attribute, it is typically complemented by CSS for complex styling and layout.

Extensible Markup Language (XML) is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable. The design goals of XML focus on simplicity, generality, and usability across the Internet. It is a textual data format with strong support via Unicode for different human languages. Although the design of XML focuses on documents, the language is widely used for the representation of arbitrary data structures such as those used in web services.

1. XML stands for extensible Markup Language
2. XML is a markup language like [HTML](#)
3. XML is designed to store and transport data
4. XML is designed to be self-descriptive

Differences between XML and HTML

XML and HTML were designed with different goals:

- XML is designed to carry data emphasizing on what type of data it is.
- HTML is designed to display data emphasizing on how data looks
- XML tags are not predefined like HTML tags.
- HTML is a markup language whereas XML provides a framework for defining markup languages.
- HTML is about displaying data, hence it is static whereas XML is about carrying information, which makes it dynamic.

HTML code for the note is given below

```
<!DOCTYPE html>
<html>
<h1>Note</h1>
<body>
<p>To:RAJ
<br>
From:RAVI
</p>
<h1>Reminder</h1>
<p>Meeting at 8am</p>
</body>
</html>
```

Output:

Note

To: Raj

From: Ravi

Reminder

Meeting at 8am.

```
<note>
  <to>Raj</to>
  <from>Ravi</from>
  <heading>Reminder</heading>
  <body>Meeting at 8am.</body>
</note>
```

Basically XML above does not do anything. XML is just information wrapped in tags. Users must require a piece of software to send, receive, store, or display it. **XML makes web development User Friendly** : Many computer systems contain data in incompatible formats. Exchanging data between incompatible systems or upgraded systems is a time-consuming task for web developers. Large amounts of data must be converted, and incompatible data is often lost. XML stores data in plain text format. This provides a software- and hardware-independent way of storing, transporting, and sharing data. **XML is Extensible**: XML applications will work as expected even if data is edited i.e. added or removed.

Example :The above note is edited into a newer version by adding date tag and hour tag , and by removing the heading tag. **Previous version**

```
<note>
  <date>2017-11-08</date>
  <hour>08:30</hour>
  <to>Raj</to>
  <from>Ravi</from>
  <body>Meeting at 8am.</body>
</note>
```

Domain Names

A domain name is a unique name that identifies a website. A **domain name** is your website **name**. A **domain name** is the address where Internet users can access your website. A **domain name** is used for finding and identifying computers on the Internet.



URL

Stands for "Uniform Resource Locator." A URL is the address of a specific webpage or file on the Internet. It is also known as web address. It is used to specify addresses on the World Wide Web. A **URL** is the fundamental network identification for any resource connected to the web.

Website

A Web site is a collection of Web pages. A website is a set of data and information about a particular subject which is available on the internet.

Web browser

A web browser is a software program that allows a user to locate, access, and display web pages. Browsers are used primarily for displaying and accessing websites on the internet.

Web Servers

- A web server is a system that delivers content or services to end users over the internet. The most simple definition is that a web server runs a website by returning HTML files over an HTTP connection.

Web Hosting

A Web host is an organization that sells or leases memory space on its servers. Web hosting is typically done in a data center, which provides services to clients that enable them to publish websites on the Internet.

Web Scripting

A scripting language is a programming language designed for integrating and communicating with other programming languages. Some of the most widely used scripting languages are JavaScript, VBScript, PHP, Perl, Python, Ruby, ASP.

Types of scripting:

Client side (VB Script, JavaScript, PHP)

Server side (ASP, JSP, PHP)

THANK YOU